Open problems:
Physical vector spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Completeness of the Lie algebra.** *Research.* Definition 1 admits pathological cases: a Hilbert space with the defining observables of a qubit ("too many vectors") and a Schwartz space with position as the only defining observable ("too few observables"). "Addressing this gap is an open problem"; the brief expects a "final definition" once it is solved. (Brief, Sec. 5, "Completeness of the Lie algebra"; Sec. 6.)

2. **Isomorphisms of quantum vector spaces.** The brief speaks of isomorphisms "in the category of quantum vector spaces" that "must preserve the Lie algebraic structure", but does not define the morphisms. (Brief, Example 3; Theorem 5, proof; Log, "Differences", last item.)

3. $\mathcal{S}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^m)$ **iff** $n = m$**.** Stated in Example 3 and argued only by "no linear transformation can change the number of defining observables", although the number of defining observables of a presentation is not shown to be an invariant. Not formalized. (Brief, Example 3 and Remark; Log, "Differences", last item.)

4. **Schwartz spaces as topological vector spaces.** "Schwartz spaces of different dimensionality are isomorphic as topological vector spaces" is stated without proof or reference. (Brief, Remark after Example 3.)

5. **Lie algebras determine physical systems.** *Research.* "Two systems are physically equivalent if and only if their Lie algebras of observables are equivalent" is asserted, and is what makes QSTO:V consistent with [B]STDIST. (Brief, Theorem 5, proof; Log, "Differences", last item.)

6. **Frame changes.** The [B]WDQ argument assumes that the defining observables X_i^A of one frame are continuous everywhere-defined observables in the other frame, that is, that a change of frame is an isomorphism of quantum vector spaces. This is not shown. (Brief, Theorem 5, proof; Log, "Differences", last item.)

7. **Ensembles on a quantum vector space.** *Research.* QSTO:V uses density operators ρ and $S(\rho) = -\operatorname{tr}(\rho \log \rho)$, and Sec. 5 calls the set of ensembles "the most natural space to consider", but no ensemble space over V is defined. In particular, whether the ensembles induce on the

pure states the quotient topology of V, under which the pure states are second countable exactly when V is, is open. (Brief, Sec. 3, QSTO:V; Sec. 5.)

8. **Field theories.** *Research.* "For a field theory, we would expect the defining observables to be at least countable"; no field-theory example of a quantum vector space is given. (Brief, Sec. 5, "Number of defining observables".)

9. **Which observables generate unitary groups.** *Research.* Since symmetric and self-adjoint coincide on V, "not all self-adjoint operators generate a unitary transformation", and "to understand which ones the integrability of the system must be checked". A characterization, analogous to complete vector fields in Hamiltonian mechanics, is not given. (Brief, Sec. 5, "Symmetric vs self-adjoint".)

From the formalization

None. Definition 1 and Examples 3 (in n dimensions) and 4 are checked in full.